

```
#include <SPI.h>

#include <MFRC522.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#define SS_PIN 10

#define RST_PIN 9

#define LED_DENIED_PIN 7

#define LED_ACCESS_PIN 6

LiquidCrystal_I2C lcd(0x3F,16,2);

MFRC522 mfrc522(SS_PIN, RST_PIN); // Instance of the class

int code[] = {32,154,149,117}; //This is the stored UID (Unlock Card)

int codeRead = 0;

String uidString;

void setup() {

    Serial.begin(9600);

    SPI.begin();    // Init SPI bus

    mfrc522.PCD_Init(); // Init MFRC522

    Serial.println("Arduino RFID reading UID");

    pinMode( LED_DENIED_PIN , OUTPUT);

    pinMode( LED_ACCESS_PIN , OUTPUT);

    lcd.init();

    // Print a message to the LCD.
```

```
    lcd.backlight();  
    lcd.setCursor(0,0);  
    lcd.print("Show your card:");  
}  
  
void loop() {  
  if ( mfrc522.PICC_IsNewCardPresent()  
    {  
      if ( mfrc522.PICC_ReadCardSerial()  
        {  
          lcd.clear();  
          Serial.print("Tag UID:");  
          lcd.setCursor(0,0);  
          lcd.print("Tag UID:");  
          lcd.setCursor(0,1);  
          for (byte i = 0; i < mfrc522.uid.size; i++) {  
              Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");  
              Serial.print(mfrc522.uid.uidByte[i], HEX);  
              lcd.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");  
              lcd.print(mfrc522.uid.uidByte[i], HEX);  
              // lcd.print(" ");  
          }  
          Serial.println();  
        }  
      }  
    }
```

```

int i = 0;

boolean match = true;

while(i<mfr522.uid.size)
{
    if(!(int(mfr522.uid.uidByte[i]) == int(code[i])))
    {
        match = false;
    }
    i++;
}

delay(3000);

lcd.clear();

lcd.setCursor(0,0);

if(match)
{
    digitalWrite( LED_ACCESS_PIN , HIGH);
    lcd.print("Authorized access");
}
else{
    digitalWrite( LED_DENIED_PIN , HIGH);
    lcd.print(" Access denied ");
    Serial.println("\nUnknown Card");
}

```

```
Serial.println("=====");

    mfrc522.PICC_HaltA();
    delay(3000);
    reset_state();
}
}
}

void reset_state()
{
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Show your card:");
    digitalWrite( LED_ACCESS_PIN , LOW);
    digitalWrite( LED_DENIED_PIN , LOW);
}
```